Curved Yang-Mills gauge theories

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Abstract*

We construct a gauge theory based on principal bundles $\mathscr{P}$ equipped with a right $\mathscr{G}$-action, where $\mathscr{G}$ is a Lie group bundle instead of a Lie group. Due to the fact that a $\mathscr{G}$-action acts fibre by fibre, pushforwards of tangent vectors via a right-translation act now only on the vertical structure of $\mathscr{P}$. Thus, we generalize pushforwards using a connection on $\mathscr{G}$ which will modify the pushforward. A horizontal distribution on $\mathscr{P}$ invariant under such a modified pushforward will provide a proper notion of Ehresmann connection. For achieving gauge invariance we impose conditions on the connection 1-form μ on $\mathscr{G}$: μ has to be a multiplicative form, *i.e.* closed w.r.t. a certain simplicial differential δ on $\mathscr{G}$, and the curvature R_μ of μ has to be δ-exact with primitive ζ; μ will be the generalization of the Maurer-Cartan form of the classical gauge theory, while the δ-exactness of R_μ will generalize the role of the Maurer-Cartan equation. This introduces the notion of multiplicative Yang-Mills connections, a connection which helped classifying singular foliations and symmetry breaking. For allowing curved connections on $\mathscr{G}$ in the dynamical theory we will need to generalize the typical definition of the curvature/field strength F on $\mathscr{P}$ by adding ζ to F.

Several examples for a gauge theory with a curved μ will be provided, including the inner group bundle of the Hopf fibration $\mathbb{S}^7 \to \mathbb{S}^4$, and a classification for gauge theories with structural simple group bundles will be provided, including a classification for whether these theories admit a classical description.

Last but not least, this paper aims to accessible for beginners which is why it only assumes knowledge of classical gauge theory.

*Abbreviations used in this paper: **LGB(s)** for Lie group bundle(s), **LAB(s)** for Lie algebra bundle(s).

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1. Introduction and summary

2. Lie group bundles with connection

We will construct a gauge theory on a principal bundle equipped with an LGB-action. Thus let us first study LGBs; since we will also generalise the Maurer-Cartan form, we will study connections on LGBs with a suitable curvature equation.

2.1. **Ehresmann connections ...**

2.1.1. *... as parallel transport.* In the following we will initially define connections via their corresponding parallel transport, for which we will assume that they are complete. That is, for a curve $\gamma\colon I \to M$ (I an open interval of $\mathbb{R}$ containing 0) we denote a parallel transport on fibre bundles $\pi_{\mathscr{F}}\colon \mathscr{F} \to M$ by

$$\begin{aligned} \mathrm{PT}_\gamma^{\mathscr{F}}\colon \mathscr{F}_{\gamma(0)} &\to \Gamma(\gamma^*\mathscr{F}), \\ p &\mapsto \mathrm{PT}_\gamma^{\mathscr{F}}(p), \end{aligned}$$

similarly for the parallel transport on $\mathscr{G}$. We expect a sort of equivariance for the parallel transport of $p \cdot g$:

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